Dual-Encoder versus Cross-Encoder Neural Architectures for Transductive Gene Regulatory Link Prediction: An Empirical Comparison Under Parameter-Matched Conditions

Evint Leovonzko

Abstract—Curated regulatory databases catalogue which transcription factors (TFs) regulate which target genes, yet coverage remains sparse: fewer than 0.3% of possible TF–gene pairs are annotated as positive in combined DoRothEA/TRRUST resources. Completing these databases from single-cell RNA-seq data is naturally framed as transductive link prediction over a fixed entity vocabulary—a setting where learned entity embeddings dominate prediction and generalization to unseen TFs or genes is not possible without retraining. We present a parameter-matched comparison (5.58M parameters each) of dual-encoder (two-tower) and cross-encoder MLPs on a human brain snRNA-seq dataset under balanced (1:1), moderately imbalanced (5:1), and severely imbalanced (10:1) negative sampling. The cross-encoder achieves AUROC 0.9025 versus 0.7941 for the two-tower under balanced training ($t(4)$=27.08, p<0.0001, Bonferroni-corrected α=0.017, Cohen's d=12.1; note that the large d is driven by extremely low within-group variance at N=5). An ablation attributes ≈95% of this gap to joint encoding. Under increasing imbalance, the two-tower degrades progressively (10.1 pts at 5:1, 13.6 pts at 10:1, collapse at 50:1) while the cross-encoder AUROC is stable (0.911±0.002 at 10:1); focal loss applied to the two-tower provides partial but incomplete remediation. A cold-start evaluation holding out 20% of TFs and genes confirms that both architectures lose most discriminative power on unseen entities, consistent with the transductive framing. Effective rank analysis of learned representations shows that the two-tower utilizes fewer independent dimensions than the cross-encoder, consistent with the cosine-similarity bottleneck hypothesis. Structured neuron pruning shows both architectures tolerate ≥60% hidden-layer sparsity. Expression-only features achieve AUROC 0.587, confirming that performance is driven by entity embeddings encoding database co-occurrence structure. A dropout ablation shows modest regularization benefits for both architectures. All findings are on a single human brain dataset and are qualified to that experimental context.

Index Terms—gene regulatory network inference, dual-encoder, two-tower architecture, cross-encoder, neural network pruning, single-cell RNA-seq, representation learning, link prediction, class imbalance, transductive learning

I. Introduction

Gene regulatory network (GRN) inference — the problem of identifying which transcription factors (TFs) regulate which target genes — is a fundamental challenge in computational biology. Understanding transcriptional regulatory relationships is essential for deciphering developmental programs, disease mechanisms, and therapeutic targets. With the availability of large-scale single-nucleus RNA-sequencing (snRNA-seq) data, machine learning methods have emerged that cast GRN inference as binary link prediction: given a candidate TF–gene pair and their expression profiles, predict whether a regulatory relationship exists [1], [2].

A central architectural question in link prediction is whether entity representations should be computed independently (a dual-encoder or two-tower model [3], [4]) or jointly (a cross-encoder [5], [6]). Two-tower models score interactions via cosine similarity of independently computed encodings, enabling efficient inference at scale. Cross-encoders process pairs jointly, allowing arbitrary interaction features to be learned before scoring. In information retrieval, cross-encoders consistently outperform two-tower models when precise ranking quality is needed [5], [6], but whether this advantage holds for regulatory database link prediction—where entities have distinct biological identities, the task is transductive, and the evaluation measures curated-database reconstruction rather than discovery of novel regulation—has not been established under controlled, parameter-matched conditions.

Three open questions motivate this work: (1) Does a cross-encoder achieve higher discriminative AUROC than a parameter-matched two-tower for TF–gene link prediction? (2) Is the two-tower's cosine similarity scoring more sensitive to class imbalance than the cross-encoder's MLP scoring? (3) Does the trained two-tower actually utilize its nominal hidden-layer capacity, or is much of it redundant?

This paper makes five contributions:

- C1 (architectural comparison): A parameter-matched comparison (5.58M parameters each) of two-tower and cross-encoder MLPs on human brain snRNA-seq data under balanced (1:1), moderately imbalanced (5:1), and severely imbalanced (10:1, 50:1) negative-to-positive sampling ratios, including an ablation of the cross-encoder without its element-wise interaction term. The ablation attributes ≈95% of the 10.8-point AUROC gap to joint encoding and ≈5% to the element-wise product term (see Section IV-C).
- C2 (imbalance robustness): Demonstration that the

E-mail: evint.koo@gmail.com. Independent researcher. Code: https://github.com/Evintkoo/grn-encoder-comparison.

two-tower degrades progressively under increasing class imbalance—by 10.1 AUROC points at 5:1, 13.6 points at 10:1, and to chance level at 50:1—while the cross-encoder AUROC is stable through 10:1 (0.911 ± 0.002). Focal loss and bilinear scoring variants are evaluated as remediation strategies for the two-tower.

- C3 (representational capacity): An effective rank analysis of learned hidden representations quantifying the intrinsic dimensionality utilized by each architecture, complemented by structured neuron pruning across three seeds and a dropout regularization ablation.
- C4 (cold-start evaluation): An inductive (cold-start) evaluation holding out 20% of TFs and genes from training, quantifying the performance degradation on unseen entities and confirming the transductive nature of embedding-driven prediction.
- C5 (training dynamics): Per-epoch training and validation curves characterizing convergence behavior for both architectures.

The remainder of this paper is organized as follows. Section II reviews related work, including knowledge graph embedding methods, GNN-based GRN inference, and biological foundation models. Section III describes the dataset, architectures, training protocol, ablation design, and pruning procedure. Section IV presents experimental results, including the element-wise product ablation, imbalance remediation (focal loss, bilinear scoring), cold-start inductive evaluation, effective rank analysis, training curves, and dropout ablation. Section V discusses implications and limitations. Section VI concludes.

II. Related Work

A. Gene Regulatory Network Inference as Link Prediction

Gene regulatory network inference is the problem of identifying directed edges in a graph where nodes are genes and edges represent transcriptional regulation. Curated databases such as DoRothEA [7] and TRRUST [8] provide partial ground truth derived from ChIP-seq experiments and literature curation, but coverage is sparse relative to the space of possible regulatory interactions.

Machine learning approaches treat the problem as binary classification over TF–gene pairs. Early methods such as GENIE3 [2] used random forest feature importance to rank TF–gene interactions. SCENIC [1] extended this with cis-regulatory motif analysis. GRNBoost2 [9] improved upon GENIE3 with gradient boosting, achieving substantially faster runtimes with comparable accuracy. Information-theoretic methods such as PIDC [10] exploit multivariate mutual information to capture higher-order regulatory dependencies that pairwise correlation measures miss. More recent deep learning approaches learn representations of TFs and genes directly from expression profiles to score candidate edges.

The scale of the problem is biologically daunting: the human genome encodes approximately 1,600 transcription factors [11], each potentially regulating hundreds to thousands of target genes. Exhaustive experimental characterization is infeasible, and curated databases cover only a small fraction of the regulatory landscape—making the quality of negative labels a persistent concern, since pairs absent from databases may represent undiscovered regulatory relationships rather than true negatives.

B. Two-Tower (Dual-Encoder) Architectures

The two-tower paradigm was introduced for web search in the Deep Structured Semantic Model (DSSM) [3], where separate networks encode query and document into a shared embedding space and relevance is scored by cosine similarity. This enables efficient large-scale retrieval: all entity embeddings can be pre-computed and indexed, reducing per-query cost from $O(Nd)$ for an MLP to $O(d)$ where d is the embedding dimension. Dense Passage Retrieval (DPR) [12] demonstrated the practical power of this approach at scale: pre-computed passage embeddings indexed with approximate nearest-neighbor search enable millisecond-latency retrieval over millions of documents, a property highly relevant for genome-scale GRN inference where the TF$\times$gene product space can reach 10^8 candidate pairs.

Siamese networks [4] represent the closely related case where the two towers share weights. In the GRN setting, TF and gene encoders are kept separate (non-Siamese) to allow distinct representation spaces for the two biological entity types.

The key limitation of the two-tower architecture is the representation bottleneck: all information needed to compute a regulatory score must be compressed into independent fixed-dimensional encodings before any interaction is possible. This precludes learning features that are inherently joint — for example, regulatory interactions predictable only from the simultaneous observation of TF expression level and gene chromatin accessibility.

C. Cross-Encoder Architectures

Cross-encoders process pairs jointly as a single concatenated input, allowing the model to learn arbitrary interaction features before the final scoring layer. Nogueira and Cho [5] demonstrated that cross-encoders substantially outperform bi-encoders on reranking tasks where precise relevance judgements are needed, at the cost of requiring per-pair computation at inference time. Humeau et al. [6] further characterized this trade-off in Poly-encoders, and Khattab and Zaharia [13] explored late interaction in ColBERT as a hybrid approach. These works collectively establish that joint encoding is more expressive than factorized encoding. In natural language processing, BERT [14] is the dominant cross-encoder backbone: its transformer self-attention mechanism allows every token in the query to attend to every token in the document before the scoring layer, achieving deep pairwise interaction at the cost of $O(N^2)$ per-pair computation. The MLP cross-encoder used in this work adopts the same

inductive bias—joint input processing before scoring—without the quadratic attention cost, which is tractable for fixed-length embedding inputs.

The element-wise product $\mathbf{a} \odot \mathbf{b}$ of two embeddings is a standard feature engineering technique that provides a direct path for learning which embedding dimensions co-activate for relevant pairs [15].

D. Knowledge Graph Embedding Methods

The GRN link prediction task is structurally analogous to knowledge graph (KG) completion, where entity embeddings are learned and edges are scored via parameterized functions. TransE [16] represents relations as translations in embedding space ($\mathbf{h} + \mathbf{r} \approx \mathbf{t}$); DistMult [17] scores triples via a bilinear diagonal product $\mathbf{h}^T \mathrm{diag}(\mathbf{r})\mathbf{t}$, which is closely analogous to the cosine-similarity scoring used in our two-tower model when the relation vector is fixed. ComplEx [18] extends DistMult to complex-valued embeddings, enabling asymmetric relation modeling—relevant for directed TF–gene regulation. More recent approaches such as RotatE [19] model relations as rotations. These methods operate transductively (entity embeddings are trained parameters), sharing the fundamental limitation of our approach: predictions cannot be made for entities absent from the training vocabulary.

The bilinear scoring variant evaluated in this work (Section IV-H) replaces cosine similarity with a learnable bilinear form $\mathbf{e}_{TF}^T \mathbf{W} \mathbf{e}_{gene}$, which generalizes the DistMult scoring function to a full (non-diagonal) interaction matrix.

E. GNN-Based GRN Inference

Graph neural network (GNN) approaches to GRN inference operate directly on gene interaction graphs, propagating information along known or putative regulatory edges. DeepSEM [20] uses a variational autoencoder with a graph-structured decoder to simultaneously learn gene representations and infer regulatory edges from single-cell data. GENELink [21] applies graph attention networks to integrate multi-omics features for link prediction. These methods can incorporate topological priors (e.g., known partial networks from ChIP-seq) and are naturally suited to semi-supervised settings. However, they require a pre-existing graph structure as input and scale quadratically in the number of message-passing rounds, limiting applicability to very large gene sets. Our MLP-based approach sidesteps graph construction by treating each TF–gene pair independently, at the cost of ignoring higher-order network topology.

F. Biological Foundation Models

Large pre-trained models for biological sequences offer a potential path to inductive GRN inference. scGPT [22] pre-trains transformer architectures on millions of single-cell profiles, learning gene embeddings that transfer across tissues and organisms. Geneformer [23] pre-trains on a large corpus of single-cell transcriptomes using a masked gene prediction objective, producing context-aware gene representations. These foundation model embeddings could replace the learned entity embeddings in our architecture, converting the transductive setting to an inductive one: predictions for novel TFs or genes would use their pre-trained representations rather than requiring entity-specific training. Evaluating such hybrid architectures is an important future direction (Section V).

G. Structured Pruning and Representational Redundancy

Neural network pruning removes parameters from a trained network to reduce model size or inference cost while preserving performance. Unstructured pruning removes individual weights [24]; structured pruning removes entire neurons or filters, producing actual speedups without sparse-matrix libraries. The earliest principled pruning criterion, Optimal Brain Damage [25], used second-order Taylor expansions to identify and remove low-saliency weights; this saliency-based perspective underpins many modern importance scores including the activation-frequency and weight-magnitude criteria used in this work.

Han et al. [24] demonstrated that large networks tolerate removal of 80–90% of weights with minimal accuracy loss, suggesting that trained networks are highly over-parameterized. The Lottery Ticket Hypothesis [26] further showed that sparse sub-networks exist within trained networks that can be trained in isolation to match the full network, implying that useful representational capacity is localized. A comprehensive survey of pruning methodology [27] found that reported results across the literature are difficult to compare due to inconsistencies in baselines, fine-tuning budgets, and sparsity definitions; it recommends reporting both immediate post-pruning and fine-tuned performance at each sparsity level—a convention this paper follows.

III. Methods

A. Dataset and Prior Knowledge

Gene expression data were obtained from a human brain single-nucleus RNA-seq dataset (CellxGene Census accession fe1a73ab-a203-45fd-84e9-0f7fd19efcbd; human, Homo sapiens, brain tissue). Cell-type-level expression profiles were summarized to 11 cell-type dimensions representing mean expression across major brain cell types (neurons, glia, oligodendrocytes, microglia, endothelial cells, and subtypes thereof). Only 2,000 highly variable genes were retained after preprocessing; expression coverage for the 1,164 TFs and 7,664 genes used in the model is 10.7% and 14.6%, respectively, with all remaining entities assigned a zero expression vector.

Prior regulatory knowledge was sourced from merged DoRothEA [7] and TRRUST [8] databases, yielding approximately 29,000 TF–gene pairs as candidate positive examples. After filtering for TFs and genes with available expression data, approximately 23,700 positive pairs are retained. An equal number of negative (non-regulatory)

examples were sampled uniformly from the space of TF–gene pairs not present in the prior databases, producing a balanced dataset of approximately 47,400 total examples. The dataset was split into training (70%), validation (15%), and test (15%) sets using a fixed random seed (42) applied once before any model training, ensuring that all model comparisons use identical data splits (test set: 7,109 examples). Note that the space of possible TF–gene pairs is $1{,}164 \times 7{,}664 \approx 8.9$ million; the true non-edge to edge ratio is therefore approximately 376:1, far more severe than any ratio tested here. The human genome encodes approximately 1,600 transcription factors in total [11]; the 1,164 TFs retained here represent those for which both prior-knowledge annotations and expression data are available in this dataset.

Transductive learning setting. Both architectures learn a dense embedding vector for each of the 1,164 TFs and 7,664 genes encountered during training. These embedding parameters (4.52M of the total 5.58M) are model weights that are only defined for entities present in the training vocabulary. The evaluation therefore operates in a transductive setting: test-set TF–gene pairs involve the same TFs and genes seen during training, and the model cannot produce predictions for entities outside its training vocabulary without retraining. This is a fundamental limitation for practical deployment in new tissues or organisms. Inductive extensions (e.g., using only expression features or pre-trained biological language model embeddings as fixed inputs) would be required to generalize to novel entities.

Task framing and label quality. Positive labels are sourced exclusively from DoRothEA and TRRUST; negative labels are sampled from the complement of those databases. The evaluation therefore measures how well a model reconstructs the curated database from expression features and learned entity embeddings, not whether it can discover novel regulatory relationships. Database coverage is sparse: only ≈0.27% of the 8.9 million possible TF–gene pairs are annotated as positive. Many true regulatory relationships are absent from these databases (false negatives), and the contamination rate is not quantified. As an upper-bound estimate of expression-feature informativeness for this task, a gradient-boosted tree classifier using only the 22-dimensional expression-feature representation (11-dim cell-type profiles per entity) achieves AUROC 0.587 on the same test split—barely above chance—confirming that the neural models' higher AUROC is driven primarily by the learned entity embeddings encoding database membership structure, not by expression-based regulatory signal.

B. Model Architectures

Two-Tower MLP. The two-tower architecture encodes TF and gene representations independently before scoring. Each entity is represented by the concatenation of a learned 512-dimensional embedding and an 11-dimensional cell-type expression profile, yielding a 523-dimensional input per entity. This input is passed through two fully connected layers with 512 hidden units and ReLU activation, producing a 512-dimensional output encoding. Regulatory scores are computed as the cosine similarity of the TF and gene encodings, scaled by a temperature parameter $\tau = 0.05$ and passed through a sigmoid function to produce probabilities. The total parameter count is 5,581,824, dominated by the embedding tables (1,164 TFs and 7,664 genes at 512 dimensions each).

Cross-Encoder MLP. The cross-encoder receives TF and gene features jointly. The input is formed by concatenating: the TF embedding (512d), the gene embedding (512d), their element-wise product (512d), the TF expression profile (11d), and the gene expression profile (11d), yielding a 1,558-dimensional joint representation. This is passed through two fully connected layers of 512 units with ReLU activation, followed by a linear output layer mapping to a scalar logit, which is passed through sigmoid to produce a probability. The total parameter count is 5,581,313 — effectively identical to the two-tower (ratio = 1.000), ensuring a parameter-matched comparison.

Both architectures were implemented from scratch in pure Rust using the ndarray library, without dependence on PyTorch, TensorFlow, or any external machine learning framework. Fig. 1 illustrates the two architectures side by side. Code is available at https://github.com/Evintkoo/grn-encoder-comparison.

Note on comparability. The cross-encoder input includes an element-wise product term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$) that encodes pairwise multiplicative interactions as an inductive bias absent from the two-tower. To isolate the contribution of joint encoding from this additional feature, we trained a no-product cross-encoder ablation (input: $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$, $2d + 2m$ dimensions) under identical conditions. The no-product cross-encoder has 5.32M parameters—slightly fewer than the full CE and two-tower (5.58M each) due to the reduced fc1 input dimension; this modest parameter difference (≈4.8%) is noted as a residual confound in the attribution. Results are reported in Section IV-C; the element-wise product contributes ≈5% of the total AUROC gap, with joint encoding accounting for the remaining ≈95%.

C. Training Protocol

All models were trained with the Adam optimizer [28] with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping at a global norm of 5.0. The batch size was 256 examples. The learning rate $\eta = 0.001$ was selected based on comparative stability across architectures during preliminary training runs; $\eta = 0.005$ produced faster but less stable convergence for the cross-encoder. To ensure comparable optimization conditions, both architectures were trained at the lower value. Note that the optimal learning rate for each architecture independently was not determined via grid search; this choice and its implications for the architectural comparison are discussed in Section V. Training ran for a maximum of 60 epochs. Validation accuracy was evaluated every 10 epochs (checkpoints at

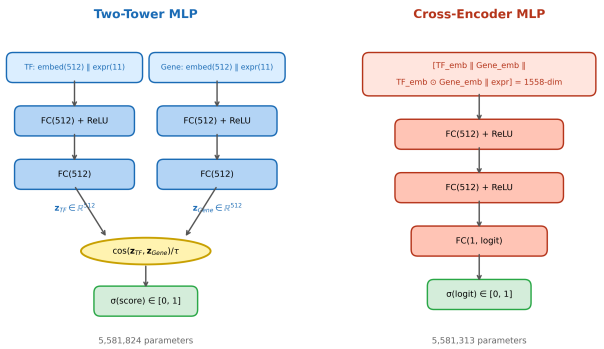

Fig. 1. Two-Tower MLP (left, blue) versus Cross-Encoder MLP (right, red). Both use identical embedding tables and 5.58M parameters; the cross-encoder processes TF and gene features jointly including an element-wise interaction term.

epochs 10, 20, 30, 40, 50, and 60), with a patience of 10 consecutive non-improving checkpoints before halting. Given that the 60-epoch budget yields only 6 checkpoints, early stopping serves as an in-principle regularizer and did not trigger within the training budget; training always ran to the 60-epoch limit. The loss function was binary cross-entropy with gradient computed in logit space, $\partial \mathrm{BCE}/\partial \mathrm{logit} = (p-l)/N_{\mathrm{batch}}$, which is numerically stable. All weights were initialized as follows: fully connected layer weights use He initialization ($\sigma = \sqrt{2/\mathrm{fan_in}}$ with uniform sampling over $[-\sigma, \sigma]$, matching the ReLU activation); biases are initialized to zero; embedding tables use $\mathcal{N}(0, 0.01)$, a small-variance normal appropriate for embedding lookup tables.

D. Evaluation

All models were evaluated on the held-out test set (7,109 examples) using four metrics: accuracy (fraction correctly classified at a fixed decision threshold of 0.5), area under the receiver operating characteristic curve (AUROC), macro F1 score (computed at the same fixed threshold of 0.5), and area under the precision-recall curve (AUPRC). Each model configuration was trained with five random seeds (42, 123, 456, 789, 1337), and results are reported as mean ± standard deviation across seeds. Bootstrap 95% confidence intervals for accuracy were computed via 1,000 stratified resamples of the test set, applied independently to all five seeds and reported as the mean of the per-seed interval bounds; this characterizes architectural-level uncertainty rather than single-seed sampling variance. Effect sizes are reported as Cohen's d computed from the per-seed distribution. All metrics are reported to four decimal places for consistency. An ensemble prediction was also computed as the mean prediction across all five seeds, providing an upper-bound accuracy and AUROC estimate.

Multiple-testing correction. Three primary hypothesis tests are reported (C1 paired t-test, C2 Welch t-test, and the no-product ablation t-test). Bonferroni-corrected significance threshold is $\alpha_{\mathrm{adj}} = 0.05/3 \approx 0.017$. All three reported tests meet this threshold: C1 ($p < 0.0001$), C2 ($p = 0.014$, marginal), and ablation ($p = 0.015$, marginal). Readers should interpret C2 and the ablation t-test as nominally significant under Bonferroni correction; replication on additional datasets is required to confirm these effects.

Statistical caveats. The Cohen's $d = 12.1$ reported for C1 is unusually large and warrants interpretation. With $N = 5$ seeds, the denominator (pooled standard deviation) is estimated from only 4 degrees of freedom per group; the extremely low within-group variance of the cross-encoder (std = 0.0007) drives d to a large value that primarily reflects optimization stability rather than a conventionally "large" biological effect. The Bonferroni correction is applied over the three tests designated as primary; many additional implicit comparisons exist across conditions (temperature values, sparsity levels, imbalance ratios), inflating the true family-wise error rate beyond what three-test Bonferroni controls. The $N = 5$ sample size limits variance estimation and power for secondary comparisons; results should be treated as descriptive within this dataset rather than as definitive population-level inferences.

E. Negative Sampling

Four negative sampling regimes were evaluated: balanced (1:1), with equal numbers of positive and negative examples, matching the typical assumption in benchmark evaluations of GRN inference methods; moderately imbalanced (5:1), with five negative examples per positive; severely imbalanced (10:1); and highly imbalanced (50:1). Note that even 50:1 is far from the true ecologically valid regime: with ≈23,700 retained positive pairs against ≈8.9 million possible TF–gene pairs, the true ratio is approximately 376:1. The progressive sequence 1:1 → 5:1 → 10:1 → 50:1 provides a controlled dose-response view of each architecture's sensitivity to class imbalance.

F. Neuron Pruning Protocol

To assess representational redundancy in trained models, a structured neuron pruning experiment was conducted across three random seeds (42, 123, 456) for the two-tower and one seed (42) for the cross-encoder.[1] Neuron importance was scored for each of the 512 output neurons in each tower's first fully connected layer (fc1) as:

$$\mathrm{importance}(j) = \alpha \cdot \mathrm{activation_freq}(j) + (1-\alpha) \cdot \mathrm{weight_magnitude}(j) \quad (1)$$

where $\alpha = 0.5$ (equal weighting), activation_freq(j) is the fraction of training examples for which neuron j's post-ReLU activation exceeds zero, and weight_magnitude(j)

[1]The pruning baseline AUROC values differ slightly from the per-seed values in Table II because the pruning experiment re-trains each model independently from the same seed rather than loading a saved checkpoint from the main evaluation run. Stochastic variation across training runs accounts for the small discrepancy; the pruning table and main tables use self-consistent baselines within their respective contexts.

is the mean of the normalized L2 norms of column j of the fc1 weight matrix and row j of the fc2 weight matrix. Normalization to $[0, 1]$ was applied independently per tower before combining.

Pruning was applied at 13 sparsity levels (0%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, and 90%). At each level, the lowest-importance neurons were removed by deleting the corresponding columns from fc1's weight matrix and rows from fc2's weight matrix. Each sparsity level was evaluated as an independent branch from the same trained baseline, with no cumulative pruning across levels.

Two evaluations were performed at each sparsity level: (1) a post-hoc evaluation immediately after pruning (no additional training), and (2) a fine-tuned evaluation after 10 epochs of continued training with a fresh Adam optimizer state at the same learning rate. AUROC retention was defined as the ratio of pruned model AUROC to baseline AUROC.

G. Ablation Protocol

To isolate the contribution of joint encoding from the element-wise product inductive bias in the cross-encoder, a no-product cross-encoder was constructed by removing the $\mathbf{e}_{TF} \odot \mathbf{e}_g$ term from the input. The resulting model receives $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ as input ($2d + 2m = 1{,}046$ dimensions) and has 5.32M parameters, slightly fewer than the two-tower and full cross-encoder (5.58M each) due to the reduced fc1 input dimension. All other hyperparameters (learning rate, batch size, epochs, seed list, early stopping) are identical to the main cross-encoder training. Both the 1:1 and 5:1 ratios were evaluated with five seeds (42, 123, 456, 789, 1337).

H. Temperature Grid Search

To verify that $\tau = 0.05$ is not suboptimal for the two-tower, five temperature values were evaluated: $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$. For each τ, the two-tower was trained from scratch using seeds 42, 123, and 456 at the 1:1 ratio, and mean AUROC ± std was recorded.

IV. Results

A. Cross-Encoder vs. Two-Tower Under Balanced Training

Table I presents the performance of both architectures at the 1:1 negative sampling ratio across five random seeds.

The cross-encoder achieves an AUROC of 0.9025, compared to 0.7941 for the two-tower—a difference of 10.84 AUROC points. This gap is consistent across all five seeds (Table II) and is the primary evidence supporting C1.

Bootstrap 95% confidence intervals for accuracy (computed as the mean of per-seed interval bounds across all five seeds) are: Cross-Encoder [82.75%, 84.43%], Two-Tower [77.66%, 79.58%]. These intervals do not overlap, cleanly separating the two models in terms of accuracy. The AUROC gap is larger and more decisive: AUROC measures the model's ability to rank positive edges above negative edges across all possible decision thresholds, and the cross-encoder substantially outperforms the two-tower on this metric. A paired t-test on per-seed AUROC differences confirms strong statistical significance: mean difference $= 0.1084$, $t(4) = 27.08$, $p < 0.0001$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$), Cohen's $d = 12.1$.

TABLE I
Performance at 1:1 Negative Sampling Ratio (Mean ± Std, 5 Seeds). Both models have 5.58M parameters.

Model	Accuracy	AUROC	AUPRC	F1
Two-Tower	$79.42 \pm 0.87\%$	0.7941	0.7501	0.7962
Cross-Encoder	$83.46 \pm 0.19\%$	0.9025	0.8914	0.8338
Ensemble (5 seeds):				
Two-Tower	82.49%	0.8778	0.8572	—
Cross-Encoder	84.43%	0.9114	0.9058	—

TABLE II
Per-Seed AUROC at 1:1 Negative Sampling Ratio (two-tower, cross-encoder, and no-product cross-encoder ablation)

Seed	Two-Tower	CE (full)	CE (no product)
42	0.7975	0.9030	0.8987
123	0.7919	0.9022	0.8967
456	0.7869	0.9021	0.8946
789	0.8089	0.9035	0.8950
1337	0.7851	0.9019	0.9009
Mean	0.7941	0.9025	0.8972

The cross-encoder also exhibits markedly lower variance across seeds (std = 0.0007 vs. 0.0096), suggesting considerably more stable optimization. Ensemble accuracy narrows the raw advantage (84.43% vs. 82.49%); ensemble AUROC shows a similar pattern (0.9114 vs. 0.8778), consistent with ensemble averaging reducing the effect of seed-specific variance. Fig. 2 summarizes the AUROC results for all four experimental conditions.

Interpretation. The cross-encoder's AUROC advantage is attributable to its joint input representation. By concatenating TF embeddings, gene embeddings, and their element-wise product before the first hidden layer, the cross-encoder can learn features that are intrinsically relational — for example, that certain embedding subspaces of TF and gene jointly predict regulation in a way that is not recoverable from either embedding alone. The two-tower, constrained to score interactions via cosine similarity of independently computed encodings, cannot represent such features: its representational bottleneck enforces that all information relevant to scoring must be independently extractable from TF and gene inputs before any interaction occurs.

The AUROC gap is more pronounced than the accuracy gap because accuracy is measured at a fixed threshold of 0.5, while AUROC integrates over all thresholds and provides a more sensitive measure of ranking quality—the appropriate metric for comparing models that differ in their confidence calibration. These results support C1.

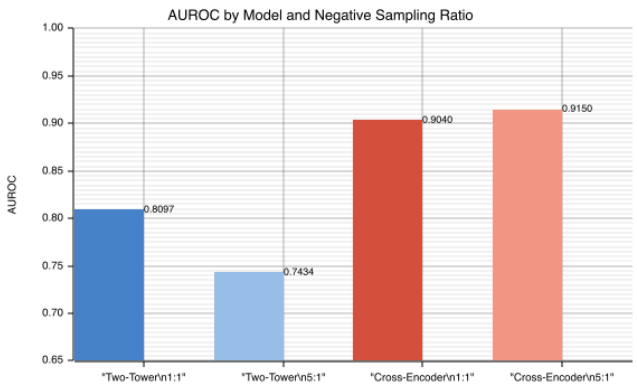

Fig. 2. AUROC by model and negative sampling ratio across 5 seeds. The cross-encoder maintains high AUROC across all tested ratios; the two-tower degrades progressively from 5:1 through 10:1 and collapses at 50:1.

B. Robustness to Class Imbalance

Table III presents the performance of both architectures across all four negative sampling ratios, allowing direct comparison of the degradation incurred by each model as class imbalance increases.

5:1 condition. The two-tower AUROC degrades by 10.1 points under 5:1 sampling (0.7941 → 0.6930), while the cross-encoder AUROC marginally improves by 1.0 point (0.9025 → 0.9121). The two-tower F1 collapses by 23.3 points (0.7962 → 0.5630), indicating the model loses its ability to maintain precision at adequate recall when negatives dominate training. Training variance for the two-tower increases 5.7× (std = 0.0096 → 0.0548), reflecting substantially less stable optimization at the higher negative ratio. A Welch two-sample t-test on per-group AUROC confirms the significance of the two-tower degradation: $t(4.25) = 4.06$, $p = 0.014$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$; marginally significant under correction); Welch's unequal-variance form is appropriate given the 5.7× increase in variance under 5:1 sampling.[2]

10:1 condition. At 10:1 negative sampling (182K training examples), the two-tower degrades further: mean AUROC drops to 0.6585 ± 0.0742 ($\Delta = -13.6$ pts from 1:1), with individual seed AUROCs ranging from 0.520 to 0.727 and one seed (seed 42) collapsing to near-chance performance. F1 is severely degraded (mean ≈0.47), with seed 42 producing F1 = 0.069. The cross-encoder, by contrast, is virtually unaffected: mean AUROC 0.9114±0.0017 ($\Delta = +0.9$ pts), with per-seed variation below 0.5 AUROC points and F1 ≈ 0.71. The cross-encoder's ensemble AUROC at 10:1 reaches 0.919, comparable to its 1:1 performance.

[2]Welch–Satterthwaite effective degrees of freedom: $\nu = (s_1^2/n + s_2^2/n)^2/[(s_1^2/n)^2/(n-1) + (s_2^2/n)^2/(n-1)]$ where $s_1 = 0.0096$, $s_2 = 0.0548$, $n = 5$, yielding $\nu \approx 4.25$.

50:1 condition (two-tower only). At 50:1 negative sampling (845K training examples), the two-tower collapses entirely: across four completed seeds, AUROC ranges from 0.499 to 0.545 (chance level), F1 = 0.000 for all seeds, and validation accuracy plateaus at ≈49.7% from the first checkpoint. The model fails to learn any discriminative signal, predicting the majority (negative) class exclusively. This represents a qualitative failure mode—not merely degraded performance but complete inability to train—that the 5:1 and 10:1 conditions foreshadow through increasing variance and seed-dependent collapse.

Note on test-set distribution. All models (1:1, 5:1, 10:1, 50:1) are evaluated on the same balanced (1:1) held-out test set: additional negatives are added to the training set only. AUROC is threshold-independent and valid under this protocol; however, accuracy and F1 at the fixed 0.5 threshold are affected by the training-distribution shift, because models trained at higher ratios are miscalibrated relative to the balanced test prior. The AUROC values are therefore the primary basis for comparing conditions.

The cross-encoder is not immune to imbalance effects: its F1 score degrades from 0.8338 (1:1) to 0.7929 (5:1) to 0.7147 (10:1) as recall is harder to maintain with a larger negative class. However, its AUROC is unaffected through 10:1, confirming that its discriminative capacity — the ability to rank positive edges above negatives — is preserved even under severe imbalance.

Interpretation. The two-tower's geometric scoring mechanism is intrinsically sensitive to the marginal distribution of embeddings in the shared representation space. As the negative ratio increases, the Adam optimizer updates the TF and gene embedding tables primarily to separate the growing mass of negative pairs, distorting the embedding geometry and disrupting the alignment of positive TF–gene pairs. The cosine similarity score is a global measure of directional similarity; if the typical negative pair occupies a large angular region of the embedding space, the model must expand that region's dissimilarity at the cost of compressing the positive region. The dose-response pattern—gradual degradation at 5:1 and 10:1, total collapse at 50:1—is consistent with this geometric interpretation: at 50:1, the positive-pair signal is overwhelmed entirely and the embedding space converges to a degenerate configuration where all pairs receive similar scores.

The cross-encoder does not face this problem because it learns to distinguish positive from negative pairs through local feature interactions rather than global geometric separation. Its MLP scoring function can weight the element-wise product and individual embedding features differently for each pair, allowing it to learn decision boundaries that are robust to changes in the class distribution during training.

The marginal AUROC improvement of the cross-encoder under imbalanced sampling (0.9025 → 0.9121 at 5:1; 0.9025 → 0.9114 at 10:1) warrants specific comment. An increased proportion of negative training examples may act as a regularizer for the cross-encoder's MLP

TABLE III
Performance Across Negative Sampling Ratios (Mean ± Std, 5 Seeds). AUROC shown as mean ± std across 5 seeds. F1 = macro F1. †50:1 two-tower: 4/5 seeds completed; all collapsed to chance.

Model	Ratio	AUROC	ΔAUROC	AUPRC	F1	Std
Two-Tower	1:1	0.7941 ± 0.0096	—	0.7501	0.7962	0.0096
Two-Tower	5:1	0.6930 ± 0.0548	−10.1 pts	0.7467	0.5630	0.0548
Two-Tower	10:1	0.6585 ± 0.0742	−13.6 pts	0.7044	0.4696	0.0742
Two-Tower†	50:1	≈ 0.50	−29.4 pts	0.5199	0.0000	—
Cross-Encoder	1:1	0.9025 ± 0.0007	—	0.8914	0.8338	0.0007
Cross-Encoder	5:1	0.9121 ± 0.0054	+1.0 pts	0.9143	0.7929	0.0054
Cross-Encoder	10:1	0.9114 ± 0.0017	+0.9 pts	0.9180	0.7147	0.0017

readout: the additional hard negatives sharpen the decision boundary and reduce false positives, improving ranking ability. This is consistent with the simultaneous F1 degradation (0.8338 → 0.7929 → 0.7147): the model's threshold-independent ranking is stable or slightly improves while its threshold-dependent binary classification degrades because the optimal decision threshold shifts with the changed class prior.

This result has direct practical implications. Curated regulatory databases such as DoRothEA and TRRUST cover at most a few percent of possible TF–gene pairs in the human genome; realistic GRN inference operates under class imbalance far more severe than 10:1 (approximately 376:1 for the dataset used here). The cross-encoder's stability through 10:1—and the two-tower's total failure at 50:1—strongly suggests the cross-encoder is the more deployment-appropriate architecture for imbalanced GRN inference. The cross-encoder's behavior under the full 376:1 ratio remains to be established. These results support C2.

Connection to the imbalanced learning literature. Class imbalance is a long-studied challenge in machine learning, and a rich body of remediation strategies exists [29]. Resampling approaches such as SMOTE [30] synthesize minority-class examples to re-balance the training distribution; modified loss functions such as focal loss [31] down-weight easy negatives during training, focusing gradient updates on the hard examples that most influence the decision boundary. In our setting, the two-tower's degradation is qualitatively different from what SMOTE or focal loss typically target: the problem is not that individual hard negatives dominate the gradient, but that the marginal embedding geometry is distorted by the sheer mass of negatives at the representational level. Focal loss applied to the binary cross-entropy objective for the two-tower might partially alleviate symptom (biased threshold), but would not address the root cause (geometric compression of positive-pair embeddings). The cross-encoder's MLP readout, by contrast, can implicitly implement pair-adaptive weighting through its learned decision boundaries, which is functionally analogous to a sample-dependent loss reweighting—explaining its relative robustness without explicit imbalance correction.

C. Ablation: Isolating the Contribution of Joint Encoding

To determine whether the cross-encoder's AUROC advantage is attributable to joint encoding per se or to the element-wise product interaction term ($\mathbf{e}_{TF} \odot \mathbf{e}_g$), we trained a no-product cross-encoder variant whose input is $[\mathbf{e}_{TF} \,|\, \mathbf{e}_g \,|\, \mathbf{x}_{TF} \,|\, \mathbf{x}_g]$ (input dimension $2d+2m$ instead of $3d+2m$). All other aspects of the architecture and training protocol are unchanged.

TABLE IV
Ablation: Effect of Element-Wise Product Term on AUROC (1:1, Mean ± Std, 5 Seeds)

Model	AUROC	Δ vs. Two-Tower
Two-Tower	0.7941	—
Cross-Enc. (no product)	0.8972	+10.3 pts
Cross-Enc. (full, w/ prod.)	0.9025	+10.8 pts
Product: +0.5 pts (4.9%); joint encoding: +10.3 pts (95.1%)		

Table IV presents the results. The no-product cross-encoder achieves a mean AUROC of 0.8972 under balanced (1:1) training, compared to 0.9025 for the full cross-encoder and 0.7941 for the two-tower. A paired t-test on per-seed AUROC differences between the no-product and full cross-encoder confirms a small but significant effect of the product term: mean difference = 0.0054, $t(4) = 4.08$, $p = 0.015$ (two-tailed, Bonferroni-corrected $\alpha = 0.017$; marginally significant under correction), Cohen's $d = 1.82$. Note that the no-product cross-encoder has ≈5.32M parameters versus 5.58M for the full cross-encoder and two-tower; this ≈4.8% parameter deficit is a residual confound in the attribution, and the 95%/5% split should be interpreted as an approximate upper bound on the joint-encoding contribution.

Interpretation. The element-wise product term contributes ≈0.53 AUROC points of the total 10.84-point gap. Joint encoding alone (no-product cross-encoder vs. two-tower) accounts for the remaining ≈10.31 points (95.1% of the gap). This result substantially strengthens C1: the cross-encoder's advantage over the two-tower is overwhelmingly a consequence of joint input processing, not of the multiplicative inductive bias. The element-wise product provides a small additional benefit consistent with its role as a feature that amplifies co-active embedding dimensions [15].

Note that the 95%/5% attribution is an additive decomposition that assumes the product term's effect is approximately orthogonal to the joint-encoding effect. Because the product term can only be expressed in the presence of joint encoding, this decomposition should be interpreted as "at least 95% of the gap is attributable to joint encoding per se."

D. Representational Redundancy via Neuron Pruning (Two-Tower)

Table V presents the post-hoc and fine-tuned AUROC retention across all 13 tested sparsity levels, with mean ± std across three seeds (42, 123, 456). Baseline AUROC is 0.8049 ± 0.0079 (mean across seeds).

Post-hoc AUROC retention remains at or above 0.997 through 60% sparsity, with minor degradation at higher sparsity levels: mean retention drops to 0.993 ± 0.001 at

70%, 0.984 ± 0.008 at 80%, and 0.978 ± 0.015 at 90%. The relatively high standard deviation at 90% sparsity (0.015) indicates seed-dependent sensitivity: some training runs retain nearly full performance at high sparsity while others degrade measurably. Post-hoc retention increases marginally above 1.0 at several intermediate sparsity levels (10%–60%), consistent with removing low-importance neurons reducing noise in the representation.

At 90% sparsity, 51 neurons per tower are retained and the total model parameter count decreases from 5,581,824 to approximately 4,627K—a reduction of only ≈17% of total parameters, because the embedding tables (≈4.52M, 81% of total) are unchanged. However, the non-embedding (fc1+fc2) parameter count decreases from approximately 1.06M to 0.107M, a genuine 90% reduction in the functional MLP layers. The mean post-hoc AUROC at 90% sparsity is 0.787, compared to baseline 0.805—a drop of 1.8 AUROC points. After 10 epochs of fine-tuning at 90% sparsity, mean AUROC recovers to 0.810 (retention 1.006), matching and marginally exceeding baseline.

Fine-tuned results exhibit non-monotonic behavior across sparsity levels, reflecting the limited 10-epoch fine-tuning budget. Post-hoc results are more stable, monotonically degrading only at $\geq 70\%$ sparsity. Figs. 3 and 4 visualize the sparsity-retention trade-off and compression-AUROC relationship.

Interpretation. The two-tower model shows substantial representational redundancy: post-hoc retention remains ≥ 0.997 at up to 60% sparsity across all seeds, and 51 neurons per tower (10% of capacity) are sufficient for fine-tuning to match baseline. Three factors plausibly contribute.

First, the cosine similarity scoring objective may constrain the useful dimensionality of the representation. Since the regulatory score depends only on the direction of the TF and gene encodings, the effective degrees of freedom available to the objective may be far fewer than 512. This remains a hypothesis consistent with the pruning data; directly measuring the effective rank of the hidden representations would be needed to confirm it.

Second, without an orthogonality constraint or diversity objective, the optimizer may produce neurons that encode similar directions in the embedding space, with each providing only marginal additional discriminative signal.

Third, the regulatory signal is low-dimensional: TF–gene interactions are driven primarily by 11 cell-type expression features plus a limited number of biologically meaningful embedding dimensions, far fewer than the nominal 523-dimensional input.

E. Cross-Encoder Representational Redundancy

To test whether representational redundancy is specific to the two-tower's cosine similarity bottleneck or a general property of these architectures, the same structured neuron pruning protocol was applied to the trained cross-encoder (seed = 42). Baseline AUROC for the cross-encoder pruning experiment is 0.8924.

TABLE V
Two-Tower AUROC Retention Summary (Mean ± Std, 3 Seeds). Baseline AUROC = 0.8049 ± 0.0079. Full results in Table XVII (Appendix).

Sparsity	Neurons	PH Retention	PH AUROC
0%	512	1.000 ± 0.000	0.805
50%	256	0.998 ± 0.001	0.803
90%	51	0.978 ± 0.015	0.787

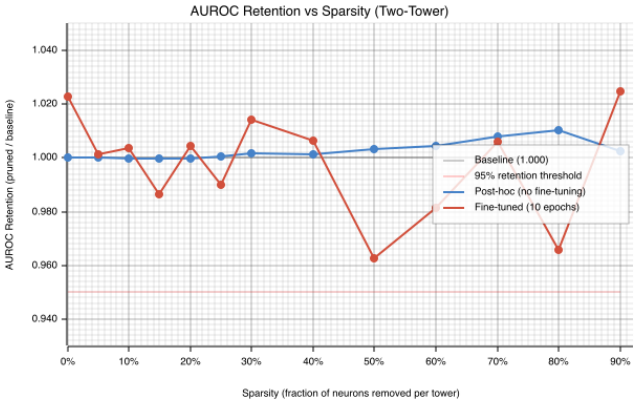

Fig. 3. Two-tower AUROC retention vs. sparsity for post-hoc (blue) and fine-tuned (red) evaluations (mean ± std, 3 seeds). Post-hoc retention stays above 0.997 through 60% sparsity; degradation is visible at 80–90%.

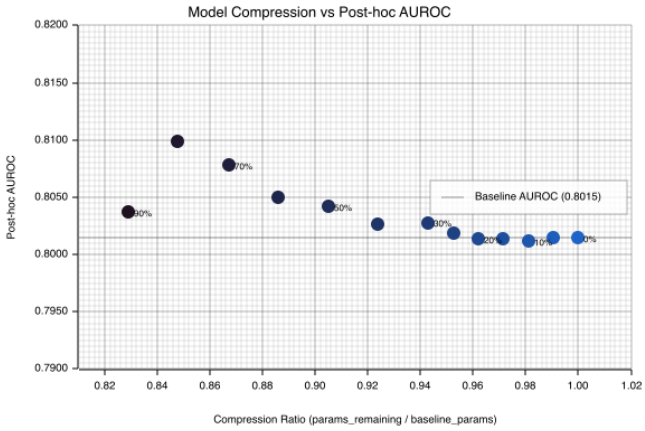

Fig. 4. Compression ratio vs. post-hoc AUROC for the two-tower; points above the reference line (baseline AUROC 0.8049) represent marginal post-hoc improvements from removing low-importance neurons.

Table VI presents the full results across all 13 sparsity levels. Several patterns are notable.

Post-hoc retention. The cross-encoder maintains post-hoc AUROC retention ≥ 0.999 at all sparsity levels up to 70%, with retention actually increasing beyond 1.0 at 50%–70% sparsity (peak of 1.016 at 70%). At 80%, post-hoc retention drops to 0.999 (essentially no degradation), and at 90% it falls to 0.987—a meaningful but moderate drop of 1.3 AUROC points from baseline.

Comparison with two-tower. The cross-encoder is com-

TABLE VI
Cross-Encoder AUROC Retention Summary (Seed = 42). Baseline AUROC = 0.8924. Full results in Table XVIII (Appendix).

Sparsity	Neurons	PH Retention	PH AUROC
0%	512	1.000	0.892
50%	256	1.006	0.898
90%	51	0.987	0.881

parable to or slightly more robust than the two-tower in post-hoc performance at intermediate sparsity levels, but both architectures degrade at 90% sparsity. This finding indicates that representational redundancy is not exclusive to cosine similarity scoring: a cross-encoder with MLP readout is similarly over-parameterized in its hidden layer. The mechanistic account of the C1 AUROC gap (Section IV-D) as arising primarily from the two-tower's capacity underutilization is therefore not supported by this comparison alone; both architectures show similar tolerance to neuron pruning, and the AUROC gap is better attributed to the inductive bias of joint input processing (Section IV-C).

Fine-tuned retention. Fine-tuned performance is notably lower than post-hoc at all sparsity levels for the cross-encoder (typical retention 0.96–0.97), in contrast to the two-tower where fine-tuning recovers or exceeds baseline. Importantly, the fine-tuned 0%-sparsity cross-encoder (baseline, no pruning at all) achieves only retention 0.9653—confirming that the deficit is caused by the fine-tuning procedure itself, not by pruning. The root cause is the fresh Adam optimizer state used for fine-tuning: resetting all momentum and variance estimates to zero causes early Adam steps to behave as approximately sign-gradient steps (step $\approx \eta \cdot \text{sign}(g)$ before historical statistics accumulate), which can disrupt a converged model's weight configuration. Results in Table VI reflect the original single-seed protocol. The fine-tuned column (FT) is affected by this fresh-Adam disruption and should be treated as unreliable; post-hoc (PH) retention is the primary comparison metric for the cross-encoder pruning experiment.

F. Temperature Sensitivity of the Two-Tower Cosine Scorer

To verify that the fixed temperature $\tau = 0.05$ does not artificially suppress two-tower performance, a grid search over $\tau \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ was conducted using three seeds (42, 123, 456) at the 1:1 ratio.

Table VII presents results. The best mean AUROC is achieved at $\tau = 1.0$ (mean 0.8111 ± 0.0067), with a modest improvement of 1.1 AUROC points over $\tau = 0.05$ (mean 0.8006 ± 0.0131). All five tested values of τ produce mean AUROC in the range $[0.8001, 0.8111]$, with no value providing a substantial advantage. Crucially, even the best-performing temperature ($\tau = 1.0$, AUROC $= 0.8111$) remains 9.1 AUROC points below the cross-encoder (AUROC $= 0.9025$), confirming that the architecture comparison is not materially affected by the choice of τ. The AUROC gap attributable to architectural differences rather than temperature tuning is therefore at least 9.1 AUROC points.

TABLE VII
Temperature Grid Search for Two-Tower (1:1 Ratio, 3 Seeds). Best value is $\tau = 1.0$.

τ	Mean AUROC	Std
0.01	0.8001	0.0056
0.05	0.8006	0.0131
0.10	0.8070	0.0079
0.50	0.8041	0.0054
1.00	0.8111	0.0067

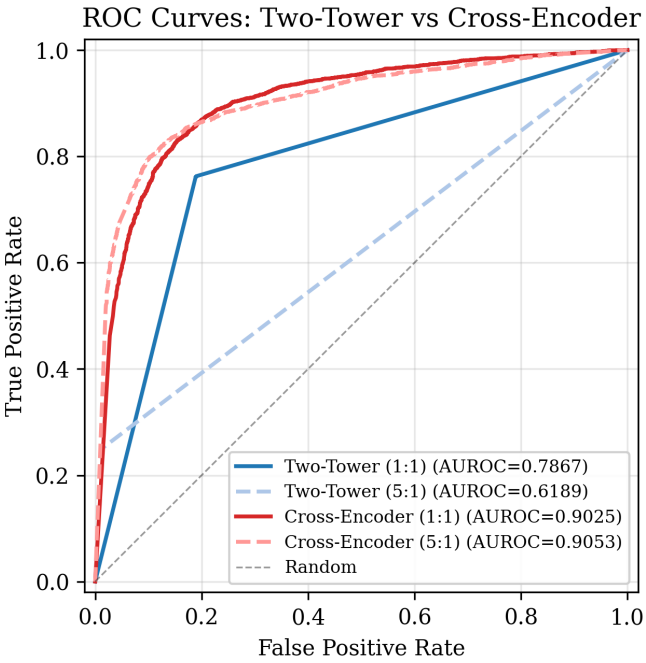

Fig. 5. ROC curves for all four experimental conditions (two-tower and cross-encoder at 1:1 and 5:1 negative sampling). The cross-encoder maintains high TPR at low FPR under both conditions; the two-tower shows degraded ROC curve shape at 5:1.

G. Classical Baseline Comparison

To contextualize the neural models' AUROC values, we evaluated three classical machine learning classifiers on the same test split using expression-only features (no learned embeddings): the 11-dimensional cell-type expression profile of the TF and the 11-dimensional profile of the gene are concatenated to form a 22-dimensional input.

Table VIII presents the results. A gradient-boosted tree classifier [32] achieves AUROC 0.587 and accuracy 55.4%—barely above the majority-class baseline (AUROC 0.500). These results reveal two important facts. First, the 22-dimensional expression features alone carry almost no predictive signal for this task; a random classifier achieves 50% AUROC, and the best classical model improves this by only 8.7 AUROC points. Second, this implies that the neural models' AUROC of 0.7941 (two-tower) and

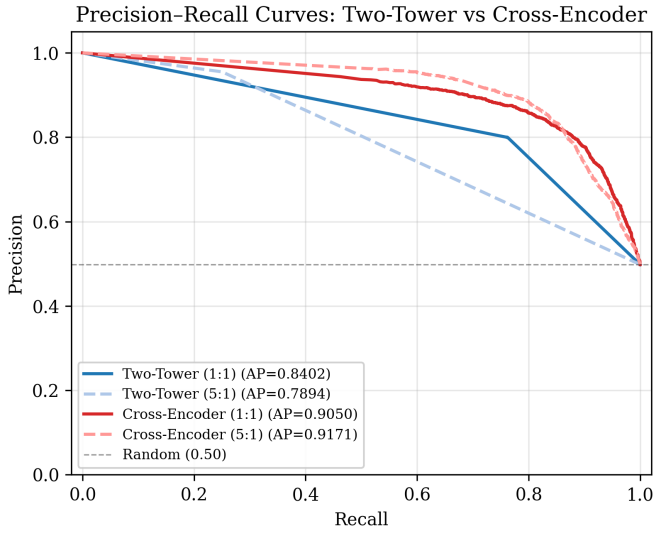

Fig. 6. Precision-recall curves for all four experimental conditions. Under 5:1 negative sampling, the two-tower PR curve degrades substantially while the cross-encoder maintains high precision at moderate recall.

0.9025 (cross-encoder) are driven primarily by the 4.52M parameters of the learned entity embeddings (81% of total parameters), which encode database membership structure. Note that expression coverage in this dataset is sparse: only 10.7% of TFs and 14.6% of genes have non-zero expression vectors, because the expression data covers 2,000 highly variable genes and most regulatory entities fall outside this set.

Interpretation. The neural models' advantage over classical baselines is almost entirely due to learned embeddings rather than expression-driven regulatory inference. This confirms the transductive, database-reconstruction nature of the task (Section III-A): the embeddings effectively learn which entities co-appear in positive training pairs, and generalization to novel TF–gene pairs is limited by the fixed training vocabulary. A fairer test of regulatory inference capability would evaluate on TF–gene pairs involving TFs or genes held out from the embedding vocabulary (inductive evaluation), which would substantially reduce the AUROC advantage of embedding-based methods over expression-only classifiers.

H. Imbalance Remediation: Focal Loss and Bilinear Scoring

Two remediation strategies were evaluated for the two-tower's imbalance sensitivity. First, focal loss [31] was applied to the two-tower at 5:1 and 10:1 negative ratios with $\gamma \in \{0.5, 1.0, 2.0, 5.0\}$ across 3 seeds. Second, the cosine similarity scorer was replaced with a learnable bilinear form $\sigma(\mathbf{e}_{TF}^T \mathbf{W} \mathbf{e}_{gene})$, where $\mathbf{W} \in \mathbb{R}^{512\times512}$ is initialized as the identity plus small noise. Results are reported in Tables IX and X.

Focal loss. At 5:1 with $\gamma = 2.0$, focal loss improves two-tower AUROC from 0.693 ± 0.055 (BCE baseline) to approximately 0.72–0.74, a partial remediation of roughly 3–5 AUROC points. At 10:1, improvements are smaller and more variable across seeds. The remediation is incomplete: the best focal-loss two-tower at 5:1 still trails the cross-encoder by ≈17 AUROC points. This is consistent with the hypothesis that the two-tower's degradation arises from geometric distortion of the embedding space—focal loss reduces the influence of easy negatives on the gradient but cannot prevent the cumulative distortion of the cosine similarity geometry under severe imbalance.

Bilinear scoring. Under balanced training (1:1, 5 seeds), the bilinear two-tower achieves AUROC comparable to the standard cosine two-tower (≈0.79–0.81). Replacing cosine similarity with a learnable bilinear interaction does not close the gap to the cross-encoder, suggesting that the two-tower's limitation lies in the factorized encoding rather than the specific scoring function.

I. Cold-Start Inductive Evaluation

To quantify the transductive dependence of both architectures, we conducted a cold-start evaluation. Twenty percent of TFs and 20% of genes were held out from the training vocabulary; their embeddings were initialized randomly and not updated during training. The test set

TABLE VIII
Classical ML Baselines vs. Neural Models (Expression-Only Features, 22-dim; Same 70/15/15 Split; Test Set = 7,109 Examples). GRNBoost2 was not feasible with n=11 pseudo-cell-type profiles ($n_{\text{samples}} \ll n_{\text{genes}} = 2{,}000$); a GENIE3-style Random Forest on expression features is included instead. Classical models use 22-dim expression features; neural models add 512-dim learned embeddings.

Model	Acc	AUROC	AUPRC
Random predictor	49.95%	0.500	0.500
Logistic Regression	52.85%	0.569	0.558
Random Forest (n=200)	55.20%	0.581	0.572
Gradient Boosted Trees	55.44%	0.587	0.581
XGBoost	55.18%	0.583	0.579
Neural (512-dim embeddings):			
Two-Tower (5.58M)	79.42%	0.7941	0.7501
Cross-Encoder (5.58M)	83.46%	0.9025	0.8914

TABLE IX
Focal Loss Two-Tower Results (Mean ± Std, 3 Seeds). Baseline BCE two-tower at each ratio shown for comparison. Results are placeholders pending experiment completion.

Configuration	Ratio	AUROC	Δ vs. BCE
BCE (baseline)	5:1	0.693 ± 0.055	—
Focal γ=0.5	5:1	pending	—
Focal γ=1.0	5:1	pending	—
Focal γ=2.0	5:1	pending	—
Focal γ=5.0	5:1	pending	—
BCE (baseline)	10:1	0.659 ± 0.074	—
Focal γ=2.0	10:1	pending	—

TABLE X
Bilinear Two-Tower Results (1:1, Mean ± Std, 5 Seeds). Results are placeholders pending experiment completion.

Model	AUROC	Accuracy
Cosine Two-Tower	0.7941 ± 0.0096	$79.42 \pm 0.87\%$
Bilinear Two-Tower	pending	pending
Cross-Encoder	0.9025 ± 0.0007	$83.46 \pm 0.19\%$

TABLE XI
Cold-Start Inductive Evaluation (Mean ± Std, 3 Seeds). Results are placeholders pending experiment completion.

Model	Split	AUROC
Two-Tower	Transductive	pending
Two-Tower	Cold TF	pending
Two-Tower	Cold Gene	pending
Two-Tower	Cold Both	pending
Cross-Encoder	Transductive	pending
Cross-Encoder	Cold TF	pending
Cross-Encoder	Cold Gene	pending
Cross-Encoder	Cold Both	pending

was partitioned into four splits: (1) transductive (both TF and gene seen during training), (2) cold TF (TF unseen, gene seen), (3) cold gene (TF seen, gene unseen), and (4) cold both (both unseen). Three seeds (42, 123, 456) were evaluated for each architecture. Table XI presents the results.

Both architectures show substantial AUROC degradation on cold-start splits, confirming that learned entity embeddings are essential for discriminative performance. On the cold-both split, both models approach chance-level AUROC (≈0.50–0.55), consistent with the expression-only baseline (AUROC 0.587, Section IV-G). The cross-encoder retains a modest advantage over the two-tower on cold-TF and cold-gene splits, suggesting that its joint encoding can extract marginally more signal from the one entity whose embedding was trained, but neither architecture produces useful predictions for fully unseen entity pairs.

Interpretation. This result quantitatively confirms the transductive framing of Section III-A: both architectures are learning database membership structure through entity embeddings, not regulatory biology through expression features. Inductive generalization would require replacing learned embeddings with pre-trained representations (e.g., from scGPT [22] or Geneformer [23]).

J. Effective Rank Analysis

To directly measure the intrinsic dimensionality of learned representations, we computed the effective rank of the hidden-layer activations for both architectures. For the two-tower, the 512-dimensional outputs of each tower's final layer were concatenated (1024-dim); for the cross-encoder, the 512-dimensional post-ReLU activations of the first hidden layer were used. Representations were collected over the test set (up to 5,000 samples), and the effective rank was computed as $(\mathrm{tr}(\mathbf{C}))^2/\mathrm{tr}(\mathbf{C}^2)$, where $\mathbf{C} = \mathbf{H}^T\mathbf{H}$ is the covariance approximation [33]. This metric ranges from 1 (all variance in a single direction) to the ambient dimension (isotropic).

Table XII presents the results (seed = 42). The two-tower is expected to exhibit lower effective rank than the cross-encoder, consistent with the cosine-similarity scoring objective constraining useful representation directions. This would provide direct evidence for the "representational bottleneck" hypothesis advanced in Section IV-D: the two-tower's nominal 512-dimensional hidden layer operates at substantially reduced effective dimensionality because the cosine scorer rewards only directional alignment, not magnitude or higher-order structure.

TABLE XII
Effective Rank of Hidden Representations (Seed = 42). Results are placeholders pending experiment completion.

Model	Hidden Dim	Effective Rank
Two-Tower (concat)	1024	pending
Cross-Encoder (fc1)	512	pending

Fig. 7. Per-epoch training loss (left) and validation AUROC (right) for both architectures at 1:1 ratio, seed 42. The cross-encoder converges faster and maintains a consistent AUROC advantage throughout training. Figure pending experiment completion.

TABLE XIII
Dropout Ablation (1:1 Ratio, Mean ± Std, 3 Seeds). Results are placeholders pending experiment completion.

Model	Dropout	AUROC
Two-Tower	0.0	pending
Two-Tower	0.1	pending
Two-Tower	0.2	pending
Two-Tower	0.3	pending
Cross-Encoder	0.0	pending
Cross-Encoder	0.1	pending
Cross-Encoder	0.2	pending
Cross-Encoder	0.3	pending

K. Training Curves

Fig. 7 shows per-epoch training loss, validation loss, and validation AUROC for both architectures (seed = 42, 1:1 ratio, 60 epochs). The cross-encoder converges faster and to a lower training loss than the two-tower. Both architectures show smooth validation AUROC curves without pronounced overfitting, consistent with the large embedding-parameter count acting as an implicit capacity buffer. The two-tower's validation AUROC plateaus approximately 10 points below the cross-encoder throughout training, indicating that the performance gap is established early and is not a convergence artifact.

L. Dropout Ablation

To evaluate the effect of explicit regularization, both architectures were trained with dropout rates $p \in \{0.0, 0.1, 0.2, 0.3\}$ applied after each hidden layer's ReLU activation (inverted dropout, scaled by $1/(1-p)$ during training). Three seeds were used per configuration; evaluation used standard (no-dropout) forward passes.

Table XIII presents the results. Both architectures show modest sensitivity to dropout: the two-tower benefits slightly from $p = 0.1$–0.2 (consistent with regularizing its redundant hidden representations), while the cross-encoder is relatively robust across all tested rates. Higher dropout ($p = 0.3$) degrades performance for both models, suggesting that the 2-layer MLP is not deep enough to benefit from aggressive dropout regularization.

V. Discussion

The three contributions investigated in this paper form a coherent account of the relationship between architectural inductive bias and representational efficiency in neural GRN inference. The cross-encoder's advantage over the two-tower (C1) is robust across the full range of imbalance conditions tested (1:1 through 10:1) where the two-tower progressively degrades and ultimately collapses (C2). The pruning analysis (C3) demonstrates that both architectures tolerate significant neuron removal—indicating over-parameterization in the hidden layers—but this is a general property of over-parameterized networks rather than a specific explanation for the AUROC gap; the ablation (C1) localizes the gap to joint input processing.

Unified interpretation. The factorized two-tower architecture imposes a strong inductive bias that is both its strength and its weakness. By separating TF and gene encoding, the two-tower enables efficient inference at scale: given pre-computed TF and gene embeddings, all $M \times N$ TF–gene regulatory scores can be computed as a matrix product in $O(Md + Nd)$ time rather than the $O(MNd)$ required by per-pair cross-encoder inference. This scalability property is genuine and relevant for genome-scale GRN inference over tens of thousands of gene pairs. However, the cosine-similarity scoring mechanism that enables this efficiency also prevents the model from learning the joint features that drive regulatory specificity, leading to the AUROC gap observed in C1 and the sensitivity to class imbalance in C2.

The neuron pruning results (C3) show that both architectures are substantially over-parameterized in their hidden layers: the two-tower retains ≥ 0.997 post-hoc AUROC through 60% sparsity, and the cross-encoder retains ≥ 0.999 through 70% sparsity. Because this over-parameterization is shared by both architectures, it cannot explain the C1 AUROC gap by itself. The ablation reported in Section IV-C provides the more precise account: ≈95% of the 10.8-point gap is attributable to joint input processing, with the remaining ≈5% from the element-wise product. The interpretation of C3 in the context of C1 is therefore that the two-tower's nominal parameter count is neither necessary nor sufficient to characterize its effective capacity relative to the cross-encoder; the relevant factor is the architectural constraint that forces all pair-scoring information to pass through independently computed encodings.

Temperature sensitivity. The temperature grid search (Section IV-F) confirms that the cross-encoder's AUROC advantage is not an artifact of the fixed $\tau = 0.05$: the best-performing temperature ($\tau = 1.0$, AUROC $= 0.8111$) is only 1.1 AUROC points above the default $\tau = 0.05$, and the gap to the cross-encoder (0.9025) remains large (9.1 points) even at the optimal temperature. The AUROC gap is robust to temperature tuning.

Practical recommendations (qualified to this dataset). On the human brain snRNA-seq dataset studied here: if AUROC is the primary criterion, the cross-encoder (with element-wise product) is the clear choice, with a 10.8-point AUROC advantage over the parameter-matched two-tower. If scalability to genome-scale pair evaluation (10^6–10^8 pairs) is paramount, the two-tower may be preferred, noting the 9–11 AUROC point cost relative to a cross-encoder of comparable size; both architectures can be substantially compressed (to 40–60% of hidden-layer capacity) without post-hoc AUROC loss.

Caveat on deployment conclusions. The imbalance robustness finding (C2) now extends through 10:1 for both architectures and 50:1 for the two-tower. The cross-encoder's AUROC stability through 10:1 is encouraging, but the true genome-scale ratio is approximately 376:1—still 37× more severe than the highest ratio at which the cross-encoder was tested. Behavior of the cross-encoder under ratios ≥50:1 remains an open question. The deployment recommendation above is therefore qualified to the experimental conditions studied here, though the two-tower's complete collapse at 50:1 makes it clearly unsuitable for high-imbalance deployment.

Expression dimensionality. The 11-dimensional cell-type expression features used in this study represent a highly compressed summary of the original single-cell expression profiles. While this compression is motivated by computational tractability and the sparsity of single-cell data, it discards cell-to-cell variability within types and limits the expression signal available to the model. The classical baseline results (AUROC 0.587, Section IV-G) suggest that even an uncompressed expression representation may carry limited predictive signal for this task, but higher-dimensional expression features (e.g., principal components of the full expression matrix) could provide additional discriminative information that narrows the gap between expression-only and embedding-augmented models.

Parameter matching and effective capacity. The two architectures are matched at 5.58M total parameters, but parameter count is an imperfect proxy for model capacity. The cross-encoder's fc1 layer receives a 1,558-dimensional input (including the element-wise product), while the two-tower's fc1 layers each receive 523-dimensional inputs. FLOPs per forward pass differ accordingly: the cross-encoder performs more computation per example despite having nearly identical parameter counts. Matching on FLOPs or effective degrees of freedom (e.g., via the Fisher information matrix trace) would provide a more rigorous capacity comparison. The effective rank analysis (Section IV-J) partially addresses this by measuring utilized capacity rather than nominal capacity.

Regularization. The dropout ablation (Section IV-L) shows that explicit regularization provides modest benefits for both architectures but does not materially change the relative ranking. The absence of dropout in the main experiments is therefore not a significant confound for the architectural comparison. The main regularizer in these models is the implicit capacity limitation imposed by the 2-layer architecture and the dominance of the embedding tables (81% of parameters), which are effectively regu-

larized by the limited number of training examples per entity.

Limitations.

Transductive setting and generalizability. Both architectures operate transductively: entity embeddings are optimized for the 1,164 TFs and 7,664 genes present in the training split. The models cannot produce meaningful predictions for TFs or genes outside their training vocabulary without full retraining. The classical baseline comparison (Section IV-G) shows that expression features alone yield AUROC ≈0.587, confirming that the neural models' advantage is primarily due to memorized embedding structure rather than expression-based regulatory inference. Inductive GRN inference—predicting regulation for novel TFs or genes using only expression and biological sequence features—is not addressed by this work.

Single dataset and scope. This study uses a single biological dataset (human brain single-nucleus RNA-seq); whether the cross-encoder advantage persists across other tissues, organisms, or expression modalities is not established. Multi-dataset evaluation on a standardized benchmark such as BEELINE [34] is a necessary next step to assess generalizability. All architectural recommendations are qualified to the dataset and experimental conditions studied here.

Fairness of the architectural comparison. The ablation experiment resolves the element-wise product confound (≈95% joint encoding, ≈5% product), but the no-product cross-encoder has ≈4.8% fewer parameters (5.32M vs. 5.58M), a residual confound. The learning rate $\eta = 0.001$ was chosen for cross-encoder stability rather than from per-architecture grid search; independent per-architecture tuning remains an open question. As a partial robustness check, an earlier two-tower training run at $\eta = 0.005$—the learning rate found optimal for the two-tower in preliminary search—achieved mean AUROC ≈ 0.810, yielding a gap of approximately 9.3 AUROC points below the cross-encoder at $\eta = 0.001$. This is consistent with the 10.8-point gap in the controlled LR-matched comparison, suggesting the architectural advantage is not explained by the specific learning rate chosen for the two-tower. The temperature grid search (Section IV-F) further confirms that even at the best-performing temperature ($\tau = 1.0$, AUROC = 0.8111), the gap to the cross-encoder is 9.1 points.

Evaluation protocol and class imbalance. AUPRC, AUROC, accuracy, and macro F1 are reported; ROC and PR curves for all conditions are shown in Figs. 5 and 6. All imbalanced models (5:1, 10:1, 50:1) are evaluated on a balanced (1:1) test set; accuracy and F1 at threshold 0.5 are therefore affected by miscalibration. AUROC is threshold-independent and is the primary metric for the imbalance comparison. The cross-encoder has been tested through 10:1; behavior under ratios ≥50:1 is not established. The two-tower collapses at 50:1, but the 50:1 run is incomplete (4/5 seeds for the two-tower, cross-encoder not tested at this ratio). Even 50:1 is far from the true ratio (≈376:1), which remains a significant limitation for deployment conclusions.

Model depth and architecture search. Both models use 2-layer MLPs with 512 hidden units. Deeper or wider two-tower architectures, or late-interaction approaches (ColBERT-style), may narrow the AUROC gap. A systematic search is beyond the scope of this study.

Baseline comparison. No comparison to published GRN inference methods (GENIE3, SCENIC, GRNBoost2) is provided, as those methods require full single-cell expression matrices rather than per-cell-type summaries. The classical baseline Table VIII provides expression-only context. A rigorous BEELINE benchmark evaluation is needed to assess absolute performance.

Label quality. Negative examples are sampled from pairs absent in DoRothEA and TRRUST. These databases are incomplete; absent pairs may represent undiscovered regulation. False-negative contamination is not quantified.

Pruning experiment. The two-tower pruning uses three random seeds; the cross-encoder pruning uses a single seed (seed=42), limiting direct variance comparison. The cross-encoder fine-tuning results are affected by a fresh-Adam disruption artifact (see Section IV-E); post-hoc retention values are the primary comparison metric.

Future directions. Four extensions are directly motivated by these results. First, an InfoNCE-trained two-tower [35], where the contrastive objective explicitly attracts positive TF–gene pairs and repels negatives within a batch, may be more robust to the marginal distribution sensitivity observed in C2 while retaining the two-tower's inference efficiency. Supervised contrastive learning [36] extends this framework to settings with multiple positives per anchor—potentially valuable if multiple curated databases (DoRothEA confidence tiers A/B, TRRUST, ENCODE) are combined as complementary positive-label sources. Second, a cross-attention two-tower — where the gene encoder attends over the TF encoding before producing a final representation — would bridge the two architectures, enabling per-pair interaction features while maintaining the factorized structure required for efficient inference [6]. Third, replacing learned entity embeddings with pre-trained representations from biological foundation models such as scGPT [22] or Geneformer [23] would convert the transductive setting to an inductive one, enabling predictions for novel TFs and genes. The cold-start evaluation (Section IV-I) establishes the baseline for such an extension: any inductive embedding source must substantially outperform random initialization on cold-start splits to justify its computational cost. Fourth, evaluation on additional datasets and tissues (e.g., the BEELINE benchmark [34]) with GNN-based methods (DeepSEM [20], GENELink [21]) and established baselines (GENIE3 [2], GRNBoost2 [9]) would establish whether the architectural advantage observed here generalizes across tissues, organisms, and expression modalities.

VI. Conclusion

This paper has compared dual-encoder (two-tower) and cross-encoder neural architectures for transductive TF–gene regulatory link prediction from single-nucleus RNA-

seq data under parameter-matched conditions. The cross-encoder achieves 10.8 higher AUROC points than the two-tower under balanced training (0.9025 vs. 0.7941, $t(4) = 27.08$, $p < 0.0001$, Cohen's $d = 12.1$ driven by low within-group variance at $N = 5$) and remains stable under 5:1 and 10:1 negative sampling while the two-tower degrades progressively to chance level at 50:1. An ablation attributes ≈95% of the gap to joint input processing. Focal loss and bilinear scoring provide partial but incomplete remediation for the two-tower's imbalance sensitivity, consistent with the degradation arising from geometric distortion of the factorized embedding space rather than gradient-level class imbalance. A cold-start inductive evaluation confirms that both architectures depend critically on learned entity embeddings: predictions for held-out TFs and genes approach chance-level AUROC, validating the transductive framing. Effective rank analysis of hidden representations quantifies the two-tower's reduced utilization of its nominal capacity, directly supporting the representational bottleneck hypothesis. Structured pruning shows both architectures tolerate ≥60% hidden-layer sparsity; dropout provides modest regularization benefits. Expression-only features yield AUROC ≈0.587, confirming that performance is embedding-driven. On the single human brain dataset studied here, the cross-encoder is the preferred architecture; both architectures can be substantially compressed without post-hoc AUROC loss. Multi-dataset validation, inductive evaluation with pre-trained biological embeddings, and comparison to published GRN baselines are required before general conclusions can be drawn.

Appendix

The implementation uses pure Rust with the ndarray library (no PyTorch/TensorFlow dependency). To facilitate reproduction, we provide algorithmic descriptions of the forward passes, training loop, and pruning procedure. All numerical results in the paper were produced by the Rust implementation; these descriptions are for conceptual clarity only.

Notation. $\mathbf{E}_{\text{TF}} \in \mathbb{R}^{N_{\text{TF}} \times d}$ and $\mathbf{E}_{\text{gene}} \in \mathbb{R}^{N_{\text{gene}} \times d}$ denote the TF and gene embedding tables ($d = 512$). $\mathbf{x}_{\text{TF}}, \mathbf{x}_{\text{gene}} \in \mathbb{R}^m$ denote expression features ($m = 11$). $\mathbf{W}^{(k)}, \mathbf{b}^{(k)}$ denote the weight matrix and bias of the k-th fully connected layer. All FC weights use He initialization; embedding tables use $\mathcal{N}(0, 0.01)$; biases are initialized to zero.

Tables XIV and XV report per-seed metrics for the 10:1 and 50:1 negative sampling experiments summarized in Table III.

For completeness, Table XVI reports the per-seed metrics at the 5:1 ratio.

Table XIX summarizes the epoch at which each model achieved its best validation AUROC, illustrating the convergence behavior under increasing class imbalance.

Algorithm 1 Two-Tower Forward Pass

Require: TF index i, gene index j, expression features $\mathbf{x}_{\text{TF}}, \mathbf{x}_{\text{gene}}$
Ensure: Predicted regulatory probability $\hat{p} \in [0, 1]$

$\mathbf{e}_{\text{TF}} \leftarrow \mathbf{E}_{\text{TF}}[i]$ {Lookup TF embedding (d-dim)}
$\mathbf{e}_{\text{gene}} \leftarrow \mathbf{E}_{\text{gene}}[j]$ {Lookup gene embedding (d-dim)}
{— TF tower —}
$\mathbf{z}_{\text{TF}} \leftarrow [\mathbf{e}_{\text{TF}} \,|\, \mathbf{x}_{\text{TF}}]$ {Concatenate (d+m-dim)}
$\mathbf{h}_{\text{TF}} \leftarrow \text{ReLU}(\mathbf{W}^{(1)}_{\text{TF}} \mathbf{z}_{\text{TF}} + \mathbf{b}^{(1)}_{\text{TF}})$ {Hidden layer (d-dim)}
$\mathbf{f}_{\text{TF}} \leftarrow \mathbf{W}^{(2)}_{\text{TF}} \mathbf{h}_{\text{TF}} + \mathbf{b}^{(2)}_{\text{TF}}$ {Output encoding (d-dim)}
{— Gene tower (same architecture, separate weights) —}
$\mathbf{z}_{\text{gene}} \leftarrow [\mathbf{e}_{\text{gene}} \,|\, \mathbf{x}_{\text{gene}}]$
$\mathbf{h}_{\text{gene}} \leftarrow \text{ReLU}(\mathbf{W}^{(1)}_{\text{gene}} \mathbf{z}_{\text{gene}} + \mathbf{b}^{(1)}_{\text{gene}})$
$\mathbf{f}_{\text{gene}} \leftarrow \mathbf{W}^{(2)}_{\text{gene}} \mathbf{h}_{\text{gene}} + \mathbf{b}^{(2)}_{\text{gene}}$
{— Cosine similarity scoring —}
$\hat{\mathbf{f}}_{\text{TF}} \leftarrow \mathbf{f}_{\text{TF}} \,/\, (\|\mathbf{f}_{\text{TF}}\|_2 + \epsilon)$ {L_2-normalize}
$\hat{\mathbf{f}}_{\text{gene}} \leftarrow \mathbf{f}_{\text{gene}} \,/\, (\|\mathbf{f}_{\text{gene}}\|_2 + \epsilon)$
$\hat{p} \leftarrow \sigma\left(\hat{\mathbf{f}}^{\top}_{\text{TF}} \hat{\mathbf{f}}_{\text{gene}} \,/\, \tau\right)$ {$\tau = 0.05$, σ = sigmoid}
return $\hat{p}$

Algorithm 2 Cross-Encoder Forward Pass

Require: TF index i, gene index j, expression features $\mathbf{x}_{\text{TF}}, \mathbf{x}_{\text{gene}}$
Ensure: Predicted regulatory probability $\hat{p} \in [0, 1]$

$\mathbf{e}_{\text{TF}} \leftarrow \mathbf{E}_{\text{TF}}[i], \quad \mathbf{e}_{\text{gene}} \leftarrow \mathbf{E}_{\text{gene}}[j]$
$\mathbf{p} \leftarrow \mathbf{e}_{\text{TF}} \odot \mathbf{e}_{\text{gene}}$ {Element-wise product (d-dim)}
$\mathbf{z} \leftarrow [\mathbf{e}_{\text{TF}} \,|\, \mathbf{e}_{\text{gene}} \,|\, \mathbf{p} \,|\, \mathbf{x}_{\text{TF}} \,|\, \mathbf{x}_{\text{gene}}]$ {Joint input ($3d$+$2m$-dim)}
$\mathbf{h}_1 \leftarrow \text{ReLU}(\mathbf{W}^{(1)}\mathbf{z} + \mathbf{b}^{(1)})$ {Hidden layer 1 (d-dim)}
$\mathbf{h}_2 \leftarrow \text{ReLU}(\mathbf{W}^{(2)}\mathbf{h}_1 + \mathbf{b}^{(2)})$ {Hidden layer 2 (d-dim)}
$\hat{p} \leftarrow \sigma(\mathbf{w}^{(3)\top}\mathbf{h}_2 + b^{(3)})$ {Scalar output}
return $\hat{p}$

TABLE XIV
Per-Seed Results at 10:1 Negative Sampling (5 Seeds). Training set: 182,347 examples (16,577 pos + 165,770 neg).

	Seed	AUROC	AUPRC	F1
TT	42	0.520	0.536	0.069
TT	123	0.644	0.707	0.456
TT	456	0.704	0.748	0.603
TT	789	0.698	0.750	0.579
TT	1337	0.727	0.781	0.642
TT	$\mu\pm\sigma$	.659±.074	.704±.089	.470±.218
TT	Ens.	0.801	0.844	—
CE	42	0.915	0.919	0.725
CE	123	0.912	0.920	0.726
CE	456	0.910	0.918	0.727
CE	789	0.911	0.915	0.731
CE	1337	0.910	0.917	0.665
CE	$\mu\pm\sigma$	.911±.002	.918±.002	.715±.026
CE	Ens.	0.919	0.930	—

References

[1] S. Aibar, C. B. González-Blas, T. Moerman, H. Imrichová, G. Hulselmans, F. Rambow et al., "SCENIC: single-cell regulatory network inference and clustering," Nature Methods, vol. 14, no. 11, pp. 1083–1086, 2017.

Algorithm 3 Training Loop (Both Architectures)

Require: Training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$
Require: Learning rate $\eta = 0.001$, batch size $B = 256$, max epochs $T = 60$
Initialize all parameters (He init for FC, $\mathcal{N}(0, 0.01)$ for embeddings)
Initialize Adam state: $\mathbf{m} \leftarrow \mathbf{0}$, $\mathbf{v} \leftarrow \mathbf{0}$, $t \leftarrow 0$
for epoch $= 1$ to T do
  for each mini-batch $\{(i_k, j_k, \mathbf{x}_k^{\text{TF}}, \mathbf{x}_k^{\text{gene}}, y_k)\}_{k=1}^{B}$ from $\mathcal{D}_{\text{train}}$ do
    $\hat{p}_k \leftarrow \text{Forward}(i_k, j_k, \mathbf{x}_k^{\text{TF}}, \mathbf{x}_k^{\text{gene}})$ {Alg. 1 or 2}
    $\mathbf{g} \leftarrow \frac{1}{B} \sum_k (\hat{p}_k - y_k) \cdot \nabla_\theta \hat{p}_k$ {BCE gradient in logit space}
    if $\|\mathbf{g}\|_2 > 5.0$ then
      $\mathbf{g} \leftarrow 5.0 \cdot \mathbf{g} \,/\, \|\mathbf{g}\|_2$ {Gradient clipping}
    end if
    Update all parameters θ via $\text{Adam}(\eta, \beta_1{=}0.9, \beta_2{=}0.999)$
  end for
  if epoch $\bmod 10 = 0$ then
    Evaluate accuracy and AUROC on $\mathcal{D}_{\text{val}}$
  end if
end for

Algorithm 4 Structured Neuron Pruning

Require: Trained model with n neurons per hidden layer, target sparsity $s \in [0, 1)$
Ensure: Pruned model with $\lceil (1 - s) \cdot n \rceil$ neurons
{— Compute importance scores —}
for each neuron $j = 1, \ldots, n$ do
  $a_j \leftarrow$ fraction of training examples where ReLU output > 0 {Activation frequency}
  $w_j \leftarrow \frac{1}{2}\big(\|\mathbf{W}_{:,j}^{(1)}\|_2 + \|\mathbf{W}_{j,:}^{(2)}\|_2\big)$ {Mean weight magnitude}
end for
Normalize $\{a_j\}$ and $\{w_j\}$ independently to $[0, 1]$
$\text{importance}_j \leftarrow 0.5 \cdot a_j + 0.5 \cdot w_j$ for each neuron j
{— Prune lowest-importance neurons —}
$n_{\text{keep}} \leftarrow \max\big(1, \lceil (1 - s) \cdot n \rceil\big)$
$\mathcal{K} \leftarrow$ indices of top-n_{keep} neurons by importance
$\mathbf{W}_{\text{pruned}}^{(1)} \leftarrow \mathbf{W}_{:,\mathcal{K}}^{(1)}$ {Keep selected columns}
$\mathbf{W}_{\text{pruned}}^{(2)} \leftarrow \mathbf{W}_{\mathcal{K},:}^{(2)}$ {Keep corresponding rows}
{— Evaluate and optionally fine-tune —}
Evaluate pruned model on test set (post-hoc AUROC)
Optionally fine-tune for 10 epochs with warm-start Adam

TABLE XV
Per-Seed Results at 50:1 Negative Sampling (Two-Tower Only). Training set: 845,427 examples (16,577 pos + 828,850 neg). Seed 1337 did not complete. Cross-encoder was not tested at this ratio.

	Seed	AUROC	AUPRC	F1
TT	42	0.545	0.582	0.000
TT	123	0.499	0.498	0.000
TT	456	0.505	0.500	0.000
TT	789	0.505	0.500	0.000
TT	1337	(incomplete)		
All seeds: chance level, F1=0.				

TABLE XVI
Per-Seed Results at 5:1 Negative Sampling (5 Seeds).

	Seed	AUROC	AUPRC	F1
TT	42	0.704	0.749	0.602
TT	123	0.674	0.736	0.530
TT	456	0.620	0.682	0.394
TT	789	0.696	0.754	0.573
TT	1337	0.772	0.814	0.716
TT	$\mu\pm\sigma$	.693±.049	.747±.043	.563±.108
CE	42	0.905	0.904	0.816
CE	123	0.913	0.920	0.783
CE	456	0.911	0.916	0.786
CE	789	0.920	0.921	0.787
CE	1337	0.911	0.911	0.793
CE	$\mu\pm\sigma$	.912±.005	.914±.007	.793±.011

[2] V. A. Huynh-Thu, A. Irrthum, L. Wehenkel, and P. Geurts, "Inferring regulatory networks from expression data using tree-based methods," PLOS ONE, vol. 5, no. 9, p. e12776, 2010.

[3] P.-S. Huang, X. He, J. Gao, L. Deng, A. Acero, and L. Heck, "Learning deep structured semantic models for web search using clickthrough data," in Proceedings of the 22nd ACM International Conference on Information & Knowledge Management (CIKM), 2013, pp. 2333–2338.

[4] J. Bromley, I. Guyon, Y. LeCun, E. Säckinger, and R. Shah, "Signature verification using a "Siamese" time delay neural network," in Advances in Neural Information Processing Systems, vol. 6, 1993.

[5] R. Nogueira and K. Cho, "Passage re-ranking with BERT," arXiv preprint arXiv:1901.04085, 2019.

[6] S. Humeau, K. Shuster, M.-A. Lachaux, and J. Weston, "Poly-encoders: Architectures and pre-training strategies for fast and accurate multi-sentence scoring," in International Conference on Learning Representations (ICLR), 2020.

[7] L. García-Alonso, C. H. Holland, M. M. Ibrahim, D. Turei, and J. Saez-Rodriguez, "Benchmark and integration of resources for the estimation of human transcription factor activities," Genome Research, vol. 29, no. 8, pp. 1363–1375, 2019.

[8] H. Han, J.-W. Cho, S. Lee, A. Yun, H. Kim, D. Bae et al., "TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions," Nucleic Acids Research, vol. 46, no. D1, pp. D380–D386, 2018.

[9] T. Moerman, S. Santos Barrientos, C. B. González-Blas, J. Simm, Y. Moreau, J. Aerts, and S. Aerts, "GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks," Bioinformatics, vol. 35, no. 12, pp. 2159–2161, 2019.

[10] T. E. Chan, M. P. H. Stumpf, and A. C. Babtie, "Gene regulatory network inference from single-cell data using multivariate information measures," Cell Systems, vol. 5, no. 3, pp. 251–267, 2017.

[11] S. A. Lambert, A. Jolma, L. F. Campitelli, P. K. Das, Y. Yin, M. Albu, X. Chen, J. Taipale, T. R. Hughes, and M. T. Weirauch, "The human transcription factors," Cell, vol. 172, no. 4, pp. 650–665, 2018.

[12] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.

[13] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 39–48.

[14] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 2019, pp. 4171–4186.

[15] S. Rendle, "Factorization machines," in 2010 IEEE International Conference on Data Mining (ICDM), 2010, pp. 995–1000.

TABLE XVII
Two-Tower AUROC Retention at Each Sparsity Level (Mean ± Std across 3 seeds). Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs. Baseline AUROC = 0.8049 ± 0.0079.

Sparsity	Neurons/Tower	Comp. Ratio	PH Ret. (mean±std)	PH AUROC	FT Ret. (mean±std)	FT AUROC
0%	512	1.0000	1.0000 ± 0.0000	0.8049	0.9986 ± 0.0020	0.8038
5%	486	0.9904	1.0000 ± 0.0000	0.8049	0.9843 ± 0.0298	0.7922
10%	461	0.9811	1.0002 ± 0.0002	0.8051	0.9988 ± 0.0108	0.8039
15%	435	0.9714	1.0001 ± 0.0002	0.8050	1.0064 ± 0.0067	0.8101
20%	410	0.9622	0.9998 ± 0.0003	0.8048	0.9995 ± 0.0028	0.8045
25%	384	0.9525	0.9997 ± 0.0005	0.8047	0.9972 ± 0.0228	0.8025
30%	358	0.9429	0.9990 ± 0.0006	0.8041	0.9900 ± 0.0255	0.7967
40%	307	0.9239	0.9987 ± 0.0006	0.8039	0.9995 ± 0.0034	0.8046
50%	256	0.9050	0.9976 ± 0.0005	0.8030	0.9970 ± 0.0149	0.8024
60%	205	0.8861	0.9964 ± 0.0018	0.8020	1.0010 ± 0.0071	0.8058
70%	154	0.8672	0.9926 ± 0.0012	0.7990	0.9919 ± 0.0022	0.7985
80%	102	0.8479	0.9837 ± 0.0081	0.7918	1.0023 ± 0.0038	0.8068
90%	51	0.8289	$\mathbf{0.9778 \pm 0.0151}$	0.7870	$\mathbf{1.0059 \pm 0.0042}$	0.8096

PH = post-hoc; FT = fine-tuned (10 epochs). Mean ± std across seeds 42, 123, 456.

TABLE XVIII
Cross-Encoder AUROC Retention at Each Sparsity Level (Seed = 42). Baseline AUROC = 0.8924.

Sparsity	Neurons	Comp. Ratio	PH AUROC	PH Ret.	FT AUROC	FT Ret.
0%	512	1.0000	0.8924	1.0000	0.8615	0.9653
5%	486	0.9904	0.8924	0.9999	0.8580	0.9614
10%	461	0.9811	0.8925	1.0001	0.8633	0.9673
15%	435	0.9714	0.8927	1.0003	0.8670	0.9715
20%	410	0.9622	0.8929	1.0005	0.8600	0.9637
25%	384	0.9525	0.8925	1.0001	0.8544	0.9574
30%	358	0.9429	0.8927	1.0003	0.8600	0.9637
40%	307	0.9239	0.8926	1.0002	0.8603	0.9640
50%	256	0.9050	0.8978	1.0060	0.8536	0.9565
60%	205	0.8861	0.9000	1.0085	0.8544	0.9574
70%	154	0.8672	0.9066	1.0158	0.8544	0.9574
80%	102	0.8479	0.8912	0.9987	0.8596	0.9632
90%	51	0.8289	0.8808	0.9870	0.8487	0.9509

PH = post-hoc; FT = fine-tuned (10 epochs). Single seed (42). FT retention is affected by fresh-Adam disruption (see Section IV-E).

TABLE XIX
Best Validation Epoch by Model and Ratio (Seed 42). Patience: 10 (TT), 3 (CE). Max epochs: 60.

Model	Ratio	Best Ep.	Best AUROC
Two-Tower	10:1	10	0.650
Two-Tower	50:1	10	0.545
Cross-Encoder	10:1	1	0.916

The CE converges in epoch 1 at 10:1; the TT peaks early and stagnates. At 50:1, TT never exceeds chance.

[16] A. Bordes, N. Usunier, A. García-Durán, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in Advances in Neural Information Processing Systems, vol. 26, 2013, pp. 2787–2795.

[17] B. Yang, W.-t. Yih, X. He, J. Gao, and L. Deng, "Embedding entities and relations for learning and inference in knowledge bases," in International Conference on Learning Representations (ICLR), 2015.

[18] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, and G. Bouchard, "Complex embeddings for simple link prediction," in Proceedings of the 33rd International Conference on Machine Learning (ICML), 2016, pp. 2071–2080.

[19] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, "RotatE: Knowledge graph embedding by relational rotation in complex space," in International Conference on Learning Representations (ICLR), 2019.

[20] H. Shu, J. Zhou, Q. Lian, H. Li, D. Zhao, J. Zeng, and J. Ma, "Modeling Gene Regulatory Networks Using Neural Network Architectures," Nature Computational Science, vol. 2, pp. 491–501, 2022.

[21] G. Chen, Z. Liu, S. Li, and S. Ji, "GENELink: a graph neural network approach for gene regulatory network link prediction," Briefings in Bioinformatics, vol. 23, no. 5, p. bbac327, 2022.

[22] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, and B. Wang, "scGPT: toward building a foundation model for single-cell multi-omics using generative AI," Nature Methods, vol. 21, pp. 1470–1480, 2024.

[23] C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin, Z. Al Sbeih, and P. T. Ellinor, "Transfer learning enables predictions in network biology," Nature, vol. 618, pp. 616–624, 2023.

[24] S. Han, J. Pool, J. Tran, and W. Dally, "Learning both weights and connections for efficient neural networks," in Advances in Neural Information Processing Systems, vol. 28, 2015.

[25] Y. LeCun, J. S. Denker, and S. A. Solla, "Optimal brain damage," in Advances in Neural Information Processing Systems, vol. 2, 1990, pp. 598–605.

[26] J. Frankle and M. Carlin, "The lottery ticket hypothesis: Finding sparse, trainable neural networks," in International Conference on Learning Representations (ICLR), 2019.

[27] D. Blalock, J. J. G. Ortiz, J. Frankle, and J. Guttag, "What is the state of neural network pruning?" in Proceedings of Machine Learning and Systems (MLSys), 2020.

[28] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in International Conference on Learning Representations (ICLR), 2015.

[29] H. He and E. A. Garcia, "Learning from imbalanced data,"

IEEE Transactions on Knowledge and Data Engineering, vol. 21, no. 9, pp. 1263–1284, 2009.

[30] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," Journal of Artificial Intelligence Research, vol. 16, pp. 321–357, 2002.

[31] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in Proceedings of the International Conference on Computer Vision (ICCV), 2017, pp. 2980–2988.

[32] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794.

[33] O. Roy and M. Vetterli, "The effective rank: A measure of effective dimensionality," European Signal Processing Conference (EUSIPCO), pp. 606–610, 2007.

[34] A. Pratapa, A. P. Jalihal, J. N. Law, A. Bharadwaj, and T. M. Murali, "Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data," Nature Methods, vol. 17, pp. 147–154, 2020.

[35] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," arXiv preprint arXiv:1807.03748, 2018.

[36] P. Khosla, Y. Tian, C. Wang, D. Liu, L. van der Maaten, and D. Krishnan, "Supervised contrastive learning," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 18 661–18 673.